

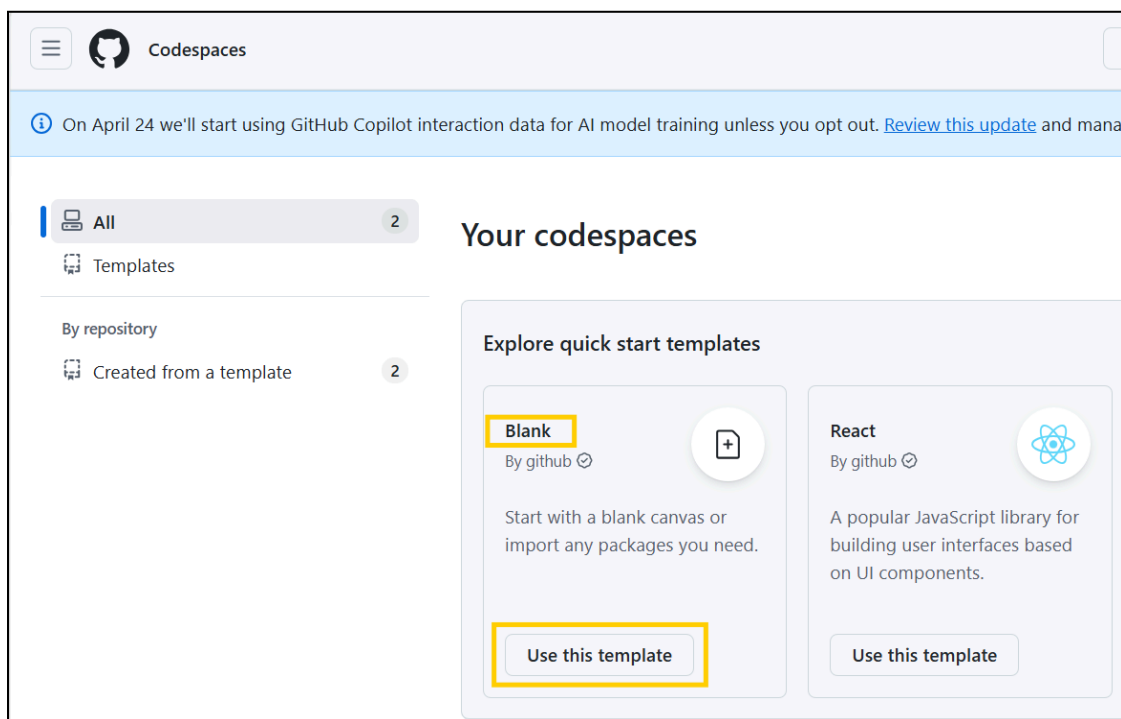
Hotel Management Agentic Application

Procedure in a nutshell

1. Download the hotel-assistant folder zip file and unzip it
2. Create a new GitHub Codespace using the Blank template
3. Create and activate virtual environment
4. Install requirements
5. In one terminal, start the backend application
6. In another terminal, start the frontend application
7. Once done, close the application and stop the codespace

Steps

1. First, create an account on [GitHub](https://github.com) if you don't have one already.
2. Then go to GitHub Codespaces: <https://github.com/codespaces>
3. Start a new Codespace using the **Blank** template

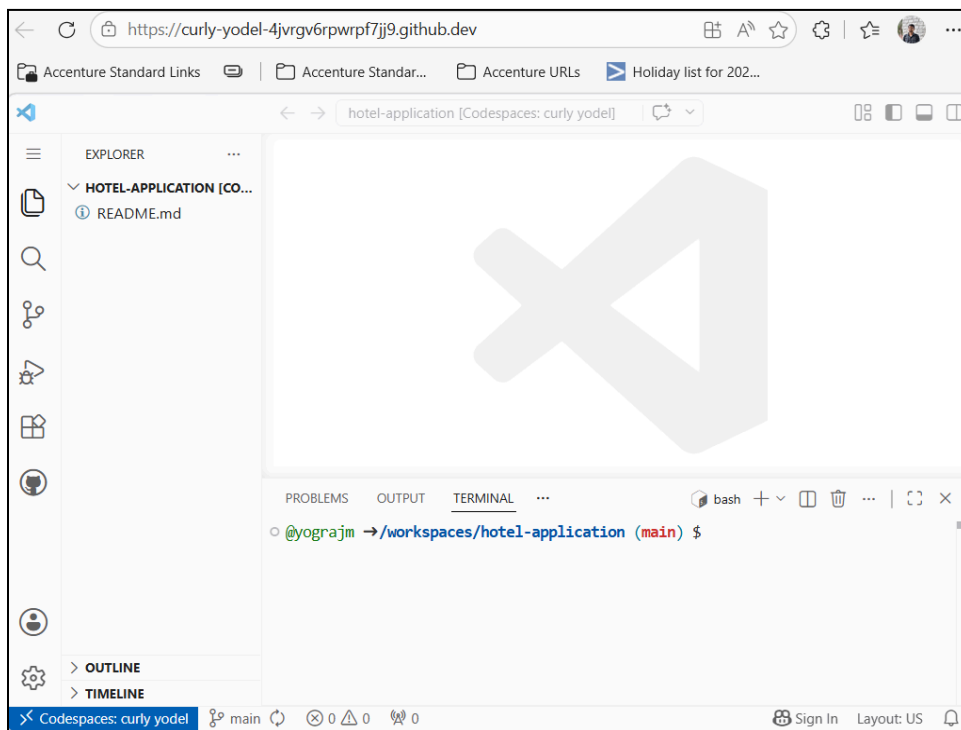


4. It will open a new tab, and start *Setting up your codespace*.

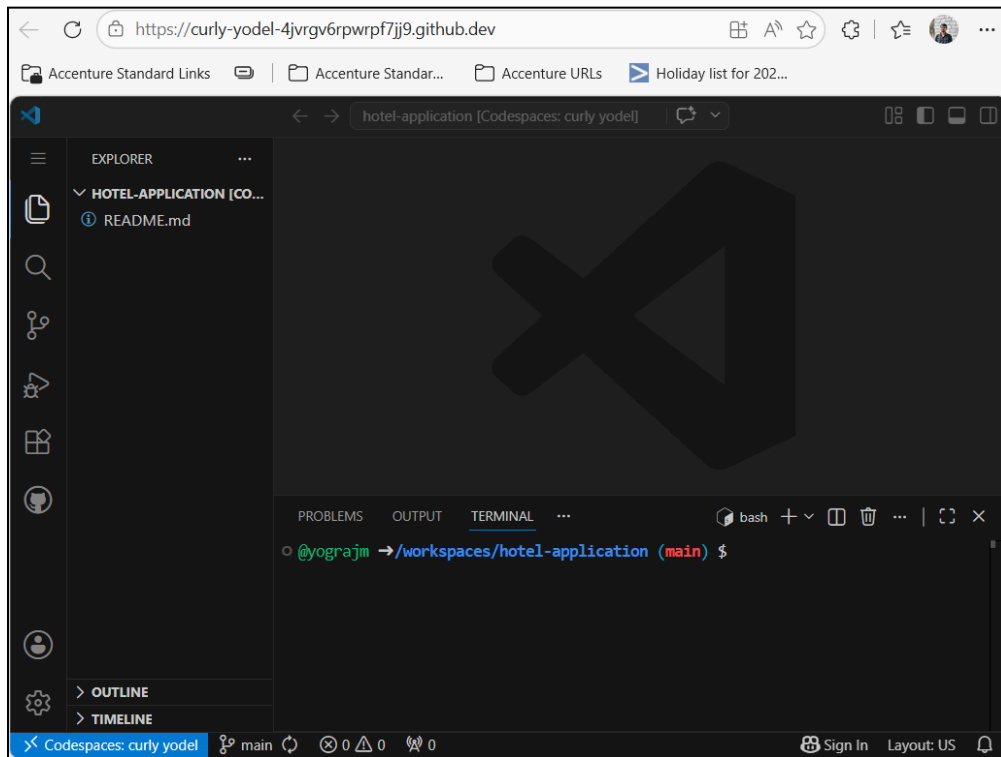
Setting up your codespace

Connecting...

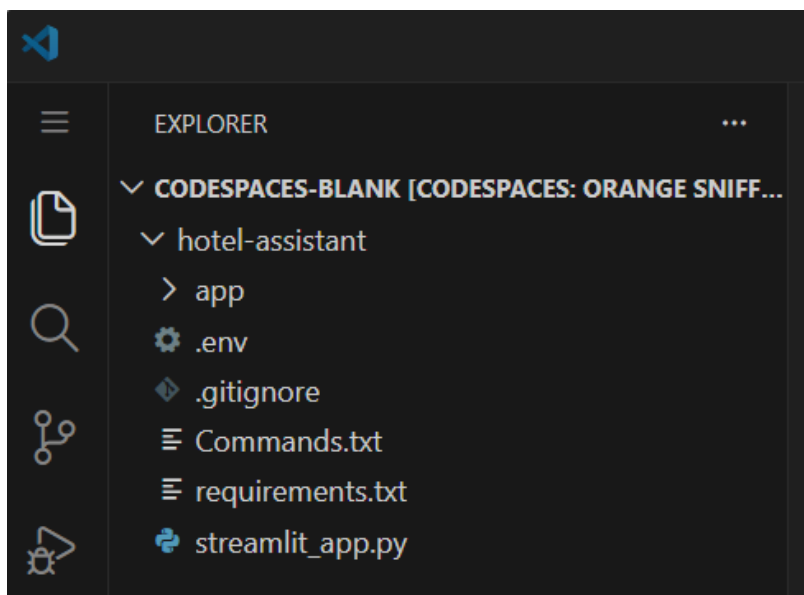
5. After loading, the Editor should open in the browser similar to below.



6. By default it is Light mode. You can change the theme by going to **Files -> Preferences -> Theme -> Color theme -> Default Dark Modern.**



7. Download the **hotel-assistant** folder zip file, shared along with this document, and unzip it in your system. Upload the entire folder to your GitHub Codespace by drag & drop.
8. Application files should be visible on the GitHub Codespace.



9. Next, create and activate a virtual environment by running the below commands:

```
python -m venv venv
```

```
source venv/bin/activate
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● @yograjm →/workspaces/codespaces-blank $ python -m venv venv
○ @yograjm →/workspaces/codespaces-blank $
● @yograjm →/workspaces/codespaces-blank $ source venv/bin/activate
○ (venv) @yograjm →/workspaces/codespaces-blank $
○ (venv) @yograjm →/workspaces/codespaces-blank $
```

10. Move into the folder and Install requirements (*This step may take some time to execute*)

```
cd hotel-assistant
```

```
pip install -r requirements.txt
```

```
● (venv) @yograjm →/workspaces/codespaces-blank $ cd hotel-assistant
○ (venv) @yograjm →/workspaces/codespaces-blank/hotel-assistant $ pip install -r requirements.txt
Collecting openai (from -r requirements.txt (line 1))
  Obtaining dependency information for openai from https://files.pythonhosted.org/packages/9d/1c/3184d72ca909bf11c9b/openai-2.36.0-py3-none-any.whl.metadata
  Downloading openai-2.36.0-py3-none-any.whl.metadata (31 kB)
Collecting langchain (from -r requirements.txt (line 2))
```

11. Review the files present in the `app/` folder.

12. Start the backend application, developed using FastAPI, with the below command:

```
uvicorn app.main:app
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS 1

○ (venv) @yograjm →/workspaces/codespaces-blank/hotel-assistant $ uvicorn app.main:app
INFO:      Started server process [8669]
INFO:      Waiting for application startup.
Hotel DB initialized.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

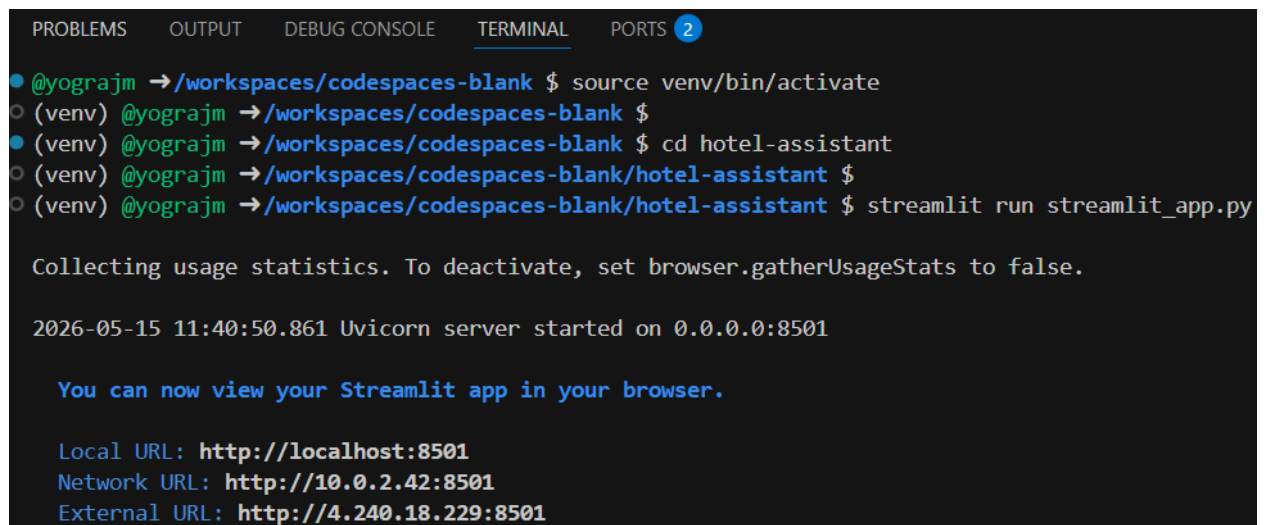
13. Review the content of the `streamlit_app.py` file.

14. Open a new Terminal by going to **More option (3 bars) >> Terminal >> New Terminal**.
15. In the new Terminal, Start the frontend application, developed using Streamlit, with the below commands:

```
source venv/bin/activate
```

```
cd hotel-assistant
```

```
streamlit run streamlit_app.py
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS 2
@yograjm →/workspaces/codespaces-blank $ source venv/bin/activate
(env) @yograjm →/workspaces/codespaces-blank $
(env) @yograjm →/workspaces/codespaces-blank $ cd hotel-assistant
(env) @yograjm →/workspaces/codespaces-blank/hotel-assistant $
(env) @yograjm →/workspaces/codespaces-blank/hotel-assistant $ streamlit run streamlit_app.py

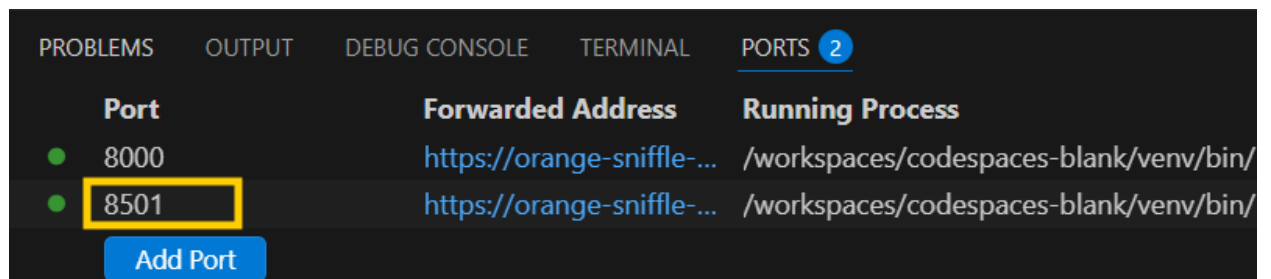
Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

2026-05-15 11:40:50.861 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.

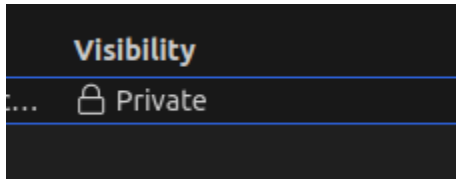
Local URL: http://localhost:8501
Network URL: http://10.0.2.42:8501
External URL: http://4.240.18.229:8501
```

16. The app should be accessible at port 8501 of localhost. You have to forward the port to access it in the browser.
Go to the **PORTS** tab beside the **TERMINAL** tab. The port should already be forwarded, else you can click on 'Add Port' and provide the port number.

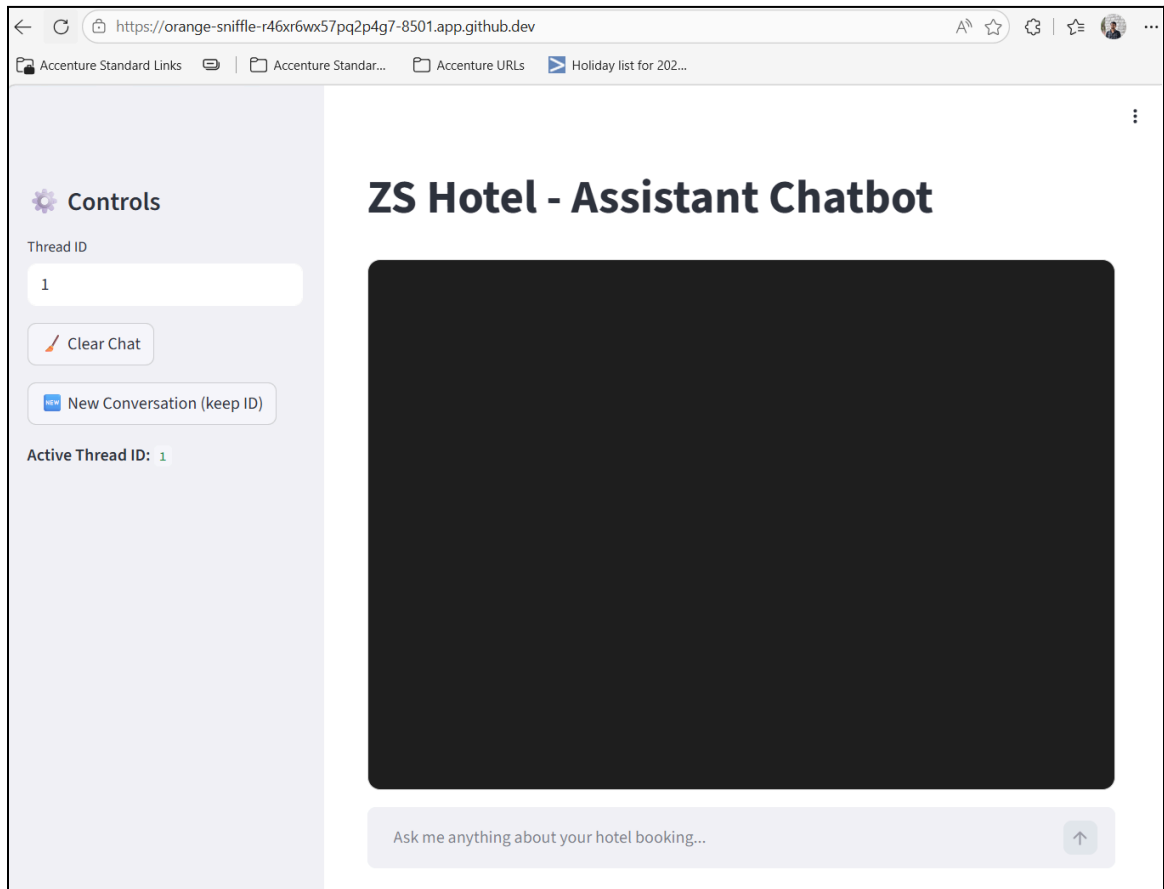


PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS 2
Port		Forwarded Address		Running Process
●	8000	https://orange-sniffle-...	/workspaces/codespaces-blank/venv/bin/	
●	8501	https://orange-sniffle-...	/workspaces/codespaces-blank/venv/bin/	
Add Port				

Similarly, you can make the port visibility to 'public' to share it with others.
Do right click on it, then **Port Visibility -> Public**.



17. Once the port is forwarded, you can access its 'Forwarded Address' to access the application.



18. Try with some user inputs:

Input: *What rooms are available?*



ZS Hotel - Assistant Chatbot

What rooms are available?

We have the following rooms available: 101, 103, 104, 105, 201, and 203. Let me know if you need more information or if you'd like to book one!

Ask me anything about your hotel booking...



Input: *Book room 105*



ZS Hotel - Assistant Chatbot

What rooms are available?

We have the following rooms available: 101, 103, 104, 105, 201, and 203. Let me know if you need more information or if you'd like to book one!

Book room 105

Could you please provide the name under which you'd like to book room 105?

Ask me anything about your hotel booking...



Input: *Bharath*

ZS Hotel - Assistant Chatbot

What rooms are available?

We have the following rooms available: 101, 103, 104, 105, 201, and 203. Let me know if you need more information or if you'd like to book one!

Book room 105

Could you please provide the name under which you'd like to book room 105?

Bharath

Room 105 is now successfully booked for Bharath. If there's anything else you need, feel free to ask!

Ask me anything about your hotel booking...

Input: *Book room 103 for Srikanth*

Input: *Book room 103 for Ramesh*

Input: *Which rooms are available?*

Input: *Book room 203 for Ramesh*

Input: *what south indian food is available*

Input: *I want to order 2 pav bhaji*

Input: *103*

Input: *Order 3 South Indian Thali to room 105*

Input: *Order 2 servings of Pasta and 1 Noodles to room 105*

Input: *I want food bill for 105*

19. Once done, close the application by pressing Ctrl + C in the terminal.

20. At last, you can stop the Codespace.

~~~END~~~